The impact of ABO blood group on the prevalence of transfusion-transmitted infections among blood donors in a tertiary-care hospital

Chilota Chibuife Efobi, MBBS, PhD[1] *Emeka Stanley Obi, MD, MHA*[2], *Oluwatobi Faniyi, MBBS*[3], *Christian Elochukwu Offiah, MBBS*[4] *Onyinyechi Victoria Okam, MD*[5], *Onyinyechukwu Joyce Ndubuisi, MBBS*[4], *Jonathan Izuchukwu Obidiegwu, MBBS*[6], *Favour Chinonso Emezue, MBBS*[4], *Okechukwu Elijah Umeh, MSc*[7]

[1]Department of Hematology and Blood Transfusion, Nnamdi Azikiwe University, Nnewi Campus, Nnewi, Nigeria; [2]Center for Cardiovascular Risk Research, East Tennessee State University, Johnson City, TN, US; [3]Lagos State University College of Medicine, Lagos, Nigeria; [4]Chukwuemeka Odumegwu Ojukwu University, College of Medical Sciences, Awka, Nigeria; [5]Babcock University Teaching Hospital, Ilishan Remo, Ogun State, Nigeria; [6]Department of Obstetrics and Gynecology, Mother of Christ Specialist Hospital, Enugu, Nigeria; [7]Medical Laboratory Department, Chukwuemeka Odumegwu Ojukwu University Teaching Hospital, Awka, Nigeria

ABSTRACT

Objectives: Transfusion-transmitted infections are a serious complication of blood transfusion. Devising a means of detecting at-risk blood donors may be beneficial in low- and middle-income countries such as Nigeria. We sought to determine the impact of ABO blood group on the prevalence of transfusion transmitted infections.

Methods: A retrospective observational study was carried out at the blood bank of Chukwuemeka Odumegwu Ojukwu University Teaching Hospital in Nigeria using data from blood donors. Information retrieved about donors included sex; blood group; and results of HIV, hepatitis B surface antigen (HBsAg), hepatitis C virus (HCV), and Venereal Disease Research Laboratory screening results. Microsoft Excel was used to sort the data and, the data analysis was conducted using R, version 4.3.2 (R Foundation for Statistical Computing). $P < .05$ was considered statistically significant.

Results: A total of 2356 donor records were reviewed; the majority of these donors were male. The prevalence of HIV, HBsAg, HCV, and venereal diseases in the study population was 0.5%, 3.3%, 1.6%, and 0.6%, respectively. There was a possible increased risk of HCV among blood group B donors ($P < .02799$).

Conclusions: Blood group B donors were found to have a higher risk of contracting HCV infection than other donors. This finding could give more insights on donor selection and screening. Further studies that are more broadly based are required to validate our findings.

KEY POINTS

- Our article identifies the prevalence of Transfusion Transmissible Infection in a tertiary hospital sited in a resource limited setting and a relationship between HCV (a TTI) and blood group B donors.
- Our article also supports a high index of suspicion in blood group B donors and HCV screening in these resource limited environments with poor access to healthcare and inefficient screening tools.
- It also shows that there are more male blood donors than females amongst other findings. This creates room for more research to find out why this is, while creating more awareness to include females.

KEY WORDS

ABO blood group; transfusion transmitted infections; blood donors; tertiary institution; blood transfusion

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Corresponding author: Onyinyechukwu Joyce Ndubuisi; missyjoyceee@gmail.com

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INTRODUCTION

Blood transfusion is a common practice in health care that has proven to be essential and lifesaving.[1] According to the World Health Organization (WHO), more than 118.5 million donated blood units are collected globally annually.[2] Blood transfusion is not without complications, however.[3] The potential for transfusion-transmitted infection (TTI) is a well-recognized complication of blood transfusions that has received considerable attention because of its transient nature and potentially lethal consequences.[4] Numerous pathogens can be transmitted through blood transfusion, including hepatitis B virus (HBV), hepatitis C virus (HCV), syphilis, and HIV.[4,5] Globally, an estimated 2 billion individuals are affected by HBV annually, with approximately 150 million people living with HCV, 12 million individuals affected by syphilis, and 33 million living with HIV.[6]

ABO is 1 of the most important classification systems, influencing blood transfusion and donation.[7] This classification system, put forward in the early 20th century by Karl Landsteiner, categorizes human blood into 4 main types—A, B, AB, and O—based on the presence or absence of specific antigens found on the surface of the red blood cells.[7,8] This grouping system is a strong predictor of the compatibility between 2 individuals, determining the transfusibility of blood between them.[9] Lethal consequences of blood transfusion have been associated with ABO system mismatches.[9] Consequently, grouping and crossmatching as well as antibody screening are indicated before every blood transfusion.[10]

Because of the risk of transmission of these infections during blood transfusions, the WHO also states that proper blood screening is an important prerequisite for blood transfusion in every center.[11] Several blood screening methods for the detection of TTIs have been developed, each operating on a basic underlying principle that enables detection of the infection.[12] Importantly, the type of blood screening method available in a particular area strongly depends on the socioeconomic dynamics of that area.[13] For instance, although rapid diagnostic testing and enzyme-linked immunosorbent assay (ELISA) are common in resource-poor countries, the financial demand associated with the application of Western blotting and nucleic acid testing makes these 2 methods a rarity in these areas.[14-16] As in many low- and middle-income countries (LMIC), in most Nigerian teaching hospitals, rapid diagnostic testing and ELISA are the most commonly used methods of screening for TTIs.[17] Although it is true that each screening technique has its unique limitations, studies have reported an increased risk of false-positive results with rapid diagnostic tests and ELISA.[17] False-positive results have been documented to occur due to cross-reactivity with other antigens or antibodies, while false-negative results are associated with an immunocompromised state and undetectability of infections during the window periods.[17] These conditions substantially limit the efficacy of these tests. In LMIC, TTIs are most commonly associated with transfusion of inadequately screened blood or blood products.[6] Importantly, TTIs contributed to declines in life expectancy among male and female individuals, making them a public health concern.[18]

Previous studies have demonstrated the relationship between the ABO blood group and host susceptibility to various pathogens, including *Helicobacter pylori, Escherichia coli, Plasmodium falciparum*, and HIV.[19] Recent studies have also suggested a possible connection between the blood groups and the occurrence of TTIs.[20,21] Tyagi et al[22] reported a possible relationship between blood group A negative and risk of contracting HIV and HBV infections. Another study demonstrated that maximum TTI seroreactivity occurred among individuals with blood group B donors (2.21%), followed closely by blood group A donors (2.16%), blood group O donors (1.57%), and AB donors (1.17%), but this finding was statistically insignificant.[21] Among donors in Ekiti State, Nigeria, Anyiam et al[23] reported that the seroprevalence of HIV was most common among blood donors with blood group O and absent in blood donors with blood groups A and AB.

Despite the existence of studies on the subject, findings have been inconclusive and inconsistent, attributable to differences in sample size, geographical and social factors, and the screening methods used. The inconsistency and inconclusiveness of findings on the relationship between TTIs and blood groups are especially notable for LMIC such as Nigeria, which have limited research work on the subject. Consequently, this study sought to bridge the research gap and contribute to the knowledge base. The main objectives of this study was to assess the correlation between TTIs and the ABO blood group and to identify any confounders that could influence that relationship. The information obtained from this study could inform the blood donor selection criteria governmental agencies develop.

METHODS

Research Design and Sites

This retrospective observational study used data obtained from the blood bank of Chukwuemeka Odumegwu Ojukwu University Teaching Hospital (COOUTH) Awka, Anambra State, Nigeria. We were able to access 2 years of data (September 2021 to February 2023) extracted from the blood bank register. Information extracted from donors included gender; blood group; and the results of HIV, hepatitis B surface antigen (HBsAg), HCV, and Venereal Disease Research Laboratory (VDRL) for syphilis screening results; data were imported into a Microsoft Excel spreadsheet.

HIV screening among blood donors was performed using the following fourth-generation ELISA-based rapid test kits: Determine (Abbott), STAT-PAK (Chembio Diagnostics, Inc), and Uni-Gold (Trinity Biotech). Determine ELISA is used as the first-line method; it is an in vitro, visually read qualitative immunoassay for the detection of antibodies to HIV-1 and HIV-2. The test employs the antigen and antibody principle to detect HIV antibodies in the plasma of the individual. Approximately 50 µL of plasma is added to the sample pad of the Determine test kit, and the plasma is absorbed through chromatography. The test is left to react for 20 minutes, then the results are read. The control line should appear for all results. If it does not appear, the results are invalid and the test should be repeated using a new test strip. A positive result on the Determine test is represented by a second line that appears after the control line. If a donor tests positive, alternative ELISA rapid test kits such as STAT-PAK and Uni-Gold are used as confirmatory tests. If the Determine result is positive and the STAT-PAK result is negative, then the Uni-Gold test is used as the tiebreaker. Global (Active Motif), ACON Labs, and Rostech ELISA

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test kits were also used to screen for HBsAg, HCV, and VDRL for syphilis infections using the same immunoassay methods.

Packed cell volume (PCV) was calculated using the HELMREASINN model JSH-120 microhematocrit centrifuge. Blood samples for PCV calculation were collected in an EDTA bottle. A nonheparinized capillary tube (Welltec) was used to collect whole-blood samples to approximately two-thirds of the length of the capillary tube. This blood is absorbed through capillary action. The end of the capillary tube from which the sample was collected is sealed with plasticine. This process is performed in duplicate, and the samples are placed opposite each other in the microhematocrit centrifuge. The centrifuge is then closed and allowed to spin 13527.80*g* for 5 minutes. The centrifuge is allowed to stop completely, and the capillary tubes are placed on the microhematocrit reader, where the PCV is read and recorded.

Inclusion Criteria

The study used the following inclusion criteria:

- Records of the blood donors who met the age criterion and actually donated blood were used for the study. Only individuals aged 18 to 65 years—the recommended age for blood donation in the COOUTH blood bank—were included in this study.
- Blood donors had to have complete data available.
- All potential blood donors, including voluntary and family replacement donors (ie, patient relatives who are called to replace units of blood used to treat a patient in case of an emergency) were included.

Exclusion Criteria

The study used the following exclusion criteria:

- Individuals with moderate or severe anemia according to WHO definitions were excluded. The cutoff PCV for blood donors in COOUTH is 37%, or 0.37l/L, as determined using a microhematocrit centrifuge.
- Donors with incomplete data were excluded.

Ethical approval was not obtained for this study.

Statistical Analysis

Data were gathered and recorded in a Microsoft Excel spreadsheet. Data analysis was conducted using R, version 4.3.2 (R Foundation for Statistical Computing) Descriptive statistics and frequency distribution were analyzed and presented in histograms, tables, and plots. Fisher exact test was used to assess the association between blood group and infection status. Logistic regression analysis was also used to assess the odd ratio of statistically significant associations between blood group and TTI.

RESULTS

A total of 2356 donor records were reviewed; the majority of donors were male. TABLE 1 lists the age, blood group, and TTI status of blood donors.

The mean (SD) PCV for included donors was 42.00 (3.40), indicating moderate variability for the average value. The median (IQR) PCV was 41 (40-44), showing that half the values fell below this point (range, 18-60).

The prevalence of HIV, HBsAg, HCV, and VDRL for syphilis infections in the study population were 0.005, 0.033, 0.016, and 0.006 respectively. The analysis of the prevalence of HIV, HBV, HCV, and syphilis across different blood groups indicated that there were no statistically significant differences for HIV, HBV, and syphilis (Fisher exact test $P = .1864$, $P = .09445$, and $P = .6442$, respectively). There was a statistically significant variation in the prevalence of HCV across blood group B, however, with $P = .02799$, as shown in TABLE 2.

TABLE 3 shows that blood group B is strongly associated with a higher likelihood of HCV positivity than blood group A. In contrast, blood groups AB and O do not show any statistically significant associations with HCV status. The findings for blood group B suggested an increased risk of HCV, while the results for blood groups AB and O indicated no clear relationship.

DISCUSSION

Blood transfusion is an essential aspect of patient treatment, especially in the developed world.[24] The process is still plagued with TTIs, however, especially in LMIC such as Nigeria, where suboptimal donor screening techniques continue to be used and a large number of donors are commercial donors, who have a higher risk of harboring TTIs.[25] Devising other means of identifying at-risk donors may be helpful in reducing the risk of TTIs in blood recipients. This study aimed to investigate the relationship between the ABO blood group and TTIs, including HIV, HCV, HBV, and syphilis, among blood donors in COOUTH, Amaku, Anambra State.

TABLE 1 Descriptive Statistics of Sex and Blood Group

Variable	Value	No. (%)
Sex	Male	2089 (88.67)
	Female	267 (11.33)
Blood group	A	389 (16.51)
	AB	9 (0.38)
	B	171 (7.26)
	O	1787 (75.85)
PCV result	No anemia	2351 (99.79)
	Anemia	5 (0.21)
HIV status	Negative	2344 (99.49)
	Positive	12 (0.51)
HBV status	Negative	2281 (96.82)
	Positive	75 (3.18)
HCV status	Negative	2319 (98.43)
	Positive	37 (1.57)
Syphilis status	Negative	2342 (99.41)
	Positive	14 (0.59)

HBV, hepatitis B virus; HCV, hepatitis C virus; PCV, packed cell volume.

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TABLE 2 Distribution of HIV, HBV, HCV, and Syphilis Infections Across Blood Groups

Infection	Blood group	Negative, No.	Positive, No.	*P* value
HIV	A	386	3	.19
	AB	9	0	
	B	169	2	
	O	1780	7	
HBV	A	383	6	.10
	AB	9	0	
	B	162	9	
	O	1727	60	
HCV	A	384	5	.03
	AB	9	0	
	B	163	8	
	O	1763	24	
Syphilis	A	386	3	.64
	AB	9	0	
	B	171	0	
	O	1776	11	

HBV, hepatitis B virus; HCV, hepatitis C virus.
*$P < .05$.

TABLE 3 Logistic Regression Analysis of HCV Status, by Blood Group

Blood group	Negative, No.	Positive, No.	*P* value	Odds ratio (95% CI)
A	384	5	[Reference]	
AB	9	0	.99	4.91×10^{-6} (4.21×10^{-10} to 4.08×10^{11})
B	163	8	.02*	3.769 ($1.239\text{-}1.264\times10^{1}$)
O	1763	24	.93	1.045 (4.30×10^{-10} to 3.12×10^{1})

*$P < .05$.

Primarily, the study sought to understand the impact of ABO blood group designations on TTI prevalence among donors.

There was a preponderance of the O blood group, and the mean PCV of the study population was 42%. The sociodemographic data of our study were in line with other studies[26] in the same region, which found similar PCVs (41.0%) to our study. The higher number of male participants is consistent with results from other studies, which show that men are more likely than women to be blood donors.[26,27] This finding may not be unrelated to personal, social, and biological factors such as menstruation, pregnancy, and lactation, that prevent women from donating blood, as evidenced in a study of blood donor characteristics in the Eastern region of Nigeria.[26,28] In addition, men are more likely to be involved in commercial blood donation in a bid to make extra money. The O blood group has consistently been shown to be most prevalent among donors across populations.[8,29]

The study participants had physiologically normal PCV ranges,[30] but a global burden of TTI as a notable complication from blood transfusion exists.[31] The prevalence of seronegative tests for the selected TTIs was high, with HIV (99.49%), HBV (96.82%), HCV (98.43%), and syphilis (99.41%) all predominantly negative among the study participants. This finding may suggest the awareness of TTIs among the blood donor group. It is also possible that the donor screening procedure is not sensitive, considering that the screening was performed using rapid test kits. Conversely, HBV had the highest percentage (3.18%) of seropositivity among the study participants, followed by HCV (1.57%). Approximately 350 million individuals worldwide have chronic HBV infection, and approximately 125 million have been infected with HCV, making HBV and HCV infection among the world's most serious infectious disease concerns.[31] Thus, prevention and early diagnosis of such infections can help reduce the infection rate and improve treatment outcomes. The prevalence rates of HIV and syphilis, at less than 1% of the study population, were slightly lower than findings from studies in the same region that assessed TTI prevalence.[26] This finding may be due to increased public health approaches in the prevention of viral diseases such as HIV and better preventive means or in a defect in the screening process, as earlier suggested.

We recorded higher numbers of HBV (60) and syphilis (11) infections among participants of group O, but this finding was not statistically significant ($P = .09$ and $P = .64$, respectively). The prevalence of TTIs among study populations can be associated with an incomplete testing of donor blood during transfusion.[32] As such, appropriate screening of donor blood can be key in preventing the life-long and sometimes fatal complications arising from TTIs.[20,31]

Research has pointed out a possible relationship between blood groups and some diseases.[22,33] There is also evidence of an association between the ABO blood group and TTIs such as HIV, HBV, HCV, and syphilis.[26,34] The results of this study indicated no statistically significant relationships between the ABO blood group and infections such as HIV, HBV, and syphilis. Although this analysis showed a higher frequency of TTIs in the O blood group, as seen in other studies,[20,21,26] there was no significant correlation ($P > .05$) that being of any particular blood group can influence positive testing on HIV, HBV, and syphilis. The higher rates of positive testing associated with the O group are related to their larger number of participants (blood group O is the most prevalent group globally). The universal donor tag associated with blood group O results in a larger number of blood donations from individuals in this group and leads to a higher number of TTIs.[8,29]

Conversely, we found a statistically significant relationship between the ABO blood group and HCV infection ($P = .02$). Participants of blood group B were observed to have a higher statistical preponderance of HCV infection than any other blood group. Although this finding is in line with a research study from Gabon,[35] other studies have produced varying results of no statistically significant relationship between the ABO blood group and any of the TTIs studied here.[20,21,32] The contrasting results can be attributed to the methodological choice adopted by the various researchers. Our findings and Ngassaki-Yoka et al[35] adopted a logistic regression analysis, which is more adept at detecting associations and adjusting for potential confounders in complex data as opposed to the contrasting findings[20,21,32] from studies that adopted a more generic use of the

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χ^2 tests, which could be less sensitive in identifying associations between variables.[36]

Despite the higher overall prevalence of TTIs among blood group O donors, our study revealed that among individuals who tested positive for HCV, individuals with blood group B had approximately 3.77 times the odds of testing positive than individuals with blood group A. This finding suggests an increased risk of HCV among blood group B individuals compared with blood group A donors. The statistically significant association with HCV in individuals with blood group B could be attributed to specific genetic or immunologic factors related to blood group antigens that influence susceptibility to HCV.[37,38]

The existence of a notable association between blood group B and HCV suggests a need for targeted screening and preventive measures for blood donors with this blood group. Such targeted screening could help identify and manage HCV infections more effectively and forestall any complications associated with this infection, including HCV-related hepatocellular carcinoma.[39]

In conclusion, we found a high prevalence of HCV infections among blood group B donors, but our study does have limitations. First, the study was limited by its retrospective nature, which prevented us from assessing the sexual behaviors of the blood donors and the presence of other high-risk behaviors. Another limitation is the lack of confirmatory tests or nucleic acid/polymerase chain reaction tests, which are usually performed on donors with reactive serological markers or rapid screening tests to confirm the presence of TTIs. These confirmatory tests are expensive and may not be available due to lack of resources in an LMIC. Confirmatory testing is done with ELISA rapid test kits, instead, which have their own limitations.

We were also unable to adequately ascertain the age range of the blood donors because some of them, while being verbally asked their ages, qualified to donate but the age was not properly documented. It would have been beneficial to know the specific age of individuals who tested positive for the various TTIs to enable adequate targeted advocacy and awareness that would mitigate the spread of TTIs in the identified age group. Further studies are required to explore genetic and immunologic factors that may explain the observed associations and to compare the high-risk behaviors of these donors based on their respective blood groups to confirm our findings.

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